

PARTH BALAR

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PROFESSIONAL EXPERIENCE

DevOps / SRE Engineer | Citta Solutions, Plano, TX

Jun 2024 – Present

- Drive reliability of production services on AWS EKS, owning SLO/SLI definitions, error budgets, and capacity/performance tuning across cloud environments.
- Champion progressive delivery: automated canary and blue-green release strategies with Terraform and GitHub Actions, cutting release cycle time by 35%.
- Built OpenTelemetry-based distributed tracing across microservices, cutting mean time-to-diagnose for cross-service latency issues by correlating traces, metrics, and logs in a single pane.
- Developed an LLM-powered on-call assistant (LangChain + Claude API) that triages PagerDuty alerts and surfaces recovery runbooks via RAG, cutting manual triage time by 50% and MTTR by 45%.
- Partnered with product and infrastructure teams to wire ML-based anomaly detection into deployment gates, blocking regressions before rollout.

Site Reliability Engineer (Contract) | Splunk, Ahmedabad, India

Jun 2022 – Dec 2023

- Owned the SLA/SLO/error-budget program for Splunk Cloud across distributed Kubernetes services on AWS and GCP, sustaining 99.9% uptime through Prometheus/Grafana alerting and Python/Bash auto-remediation.
- Built zero-downtime CI/CD pipelines (Jenkins, Terraform) and authored reusable Terraform modules adopted as the team onboarding standard, cutting deployment cycle time by 40% and provisioning time by 60%.
- Re-architected indexing/ingestion pipelines over HTTPS/TLS, cutting query latency 30% and ingestion overhead 20%; surfaced regressions on dashboards before user impact.
- Served in the 24x7 on-call rotation, driving incident response end-to-end from alert triage through mitigation, escalation, and customer-impact communication.
- Led blameless postmortems and chaos-engineering reviews; mentored junior engineers and codified findings into automated recovery runbooks, cutting MTTR by 40%.

Site Reliability Engineer Intern | Crest Data, Ahmedabad, India

Dec 2021 – May 2022

- Containerized legacy services with Docker onto Kubernetes (AWS); automated provisioning with Terraform and Jenkins, cutting deployment setup from hours to minutes.
- Helped establish the team's first SLI/SLO dashboards (Grafana, Splunk) and postmortem process, contributing to a 35% MTTR reduction in the first quarter.
- Built a Python (FastAPI) + React automation tool replacing manual data-processing workflows: 60% less manual effort, 40% fewer data errors.

PROJECTS

Berthline: AI Agent Governance Control Plane · Python, FastAPI, Kubernetes (Kopf), PostgreSQL, Redis, Helm, React/TypeScript · [Demo Video](#)

- Architected a self-hostable control plane that sits between AI agent code and production, enforcing isolation, policy, approvals, release gates, and signed audit for every agent and model; deployable in-VPC via Helm.
- Designed a Kubernetes-native runtime (Kopf operator) that runs untrusted agents in hardened Jobs: non-root, read-only root filesystem, all capabilities dropped, no service-account token, default-deny NetworkPolicy egress.
- Built fail-closed governance where every gate defaults to deny: automated PII/secret scanning and coverage enforcement for datasets, plus model promotion as a CI/CD release gate with two-person approval and separation of duties.
- Engineered tamper-evident audit: Ed25519-signed, hash-chained records with verifiable evidence packs and a full dataset→model→agent→action chain of custody; per-tenant envelope encryption (AES-GCM, KMS) blocks cross-tenant access.
- Shipped Mission Control, a real-time React/TypeScript operations dashboard (TanStack Query, React Flow lineage graphs) covering runs, approvals, model registry, and audit; instrumented end-to-end with OpenTelemetry and Prometheus.

Graph-Enhanced RAG Retrieval System · Python, sentence-transformers, ChromaDB

- Built a hybrid retrieval pipeline combining dense vector search (MiniLM embeddings) with BFS knowledge-graph traversal, surfacing structurally related code invisible to keyword search; pre-computed L2-normalized embeddings enable single matrix-multiply scoring across 80+ entities.

• TECHNICAL SKILLS

- **Cloud & Containers:** AWS (EC2, EKS, RDS, VPC, IAM, S3, CloudWatch), GCP, Kubernetes, Docker, Helm
- **IaC & CI/CD:** Terraform, Ansible, GitHub Actions, Jenkins, GitLab CI/CD
- **Observability:** Prometheus, Grafana, OpenTelemetry, Splunk, CloudWatch, ELK, PagerDuty
- **Programming:** Python, Bash, SQL, TypeScript, JavaScript, React, Java, C/C++
- **Data & AI:** PostgreSQL, MySQL, Redis, MongoDB, LangChain, LangGraph, RAG, ChromaDB, CrewAI, AutoGen

EDUCATION

M.S. Information Technology Management | The University of Texas at Dallas

Aug 2023 – May 2025

B.S. Information Technology | Dharmsinh Desai University, India

Jun 2018 – May 2022